



2011 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 2

Monday 28th February 2011

General Instructions

- Writing time — 2 Hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Structure of the paper

- Total marks — 78
- All six questions may be attempted.
- All six questions are of equal value.

Collection

- Write your candidate number clearly on each booklet.
- Hand in the six questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place the question paper inside your answer booklet for Question One.

Checklist

- SGS booklets — 6 per boy
- Candidature — 87 boys

Examiner
TCW

QUESTION ONE (13 marks) Use a separate writing booklet.**Marks**

(a) Solve $z^2 + 4z + 5 = 0$.

2(b) Given $z = 2 - i$ and $w = 3 - 4i$, express the following in the form $a + ib$, where a and b are real:

(i) iz

1

(ii) $\overline{z + w}$

1

(iii) $\frac{1}{w}$

1

(iv) $(z - 1)^2$

1(c) Given that $5 + 2i$ is a root of the equation $z^2 - 10z + n = 0$, find n .**2**(d) (i) Find the complex square roots of $-24 + 10i$.**3**

(ii) Hence, or otherwise, solve:

2

$$z^2 - (3 + i)z + 8 - i = 0$$

QUESTION TWO (13 marks) Use a separate writing booklet.**Marks**

(a) Find:

(i) $\int \frac{x^2}{(x^3 + 1)^2} dx$

2

(ii) $\int \frac{1}{\sqrt{16 - 9x^2}} dx$

2

(b) Evaluate:

(i) $\int_0^{\frac{\pi}{6}} \sec 2x \tan 2x dx$

2

(ii) $\int_{\frac{\pi}{3}}^{\pi} \frac{\sin x}{1 - \cos x} dx$

2(c) Use integration by parts to find $\int x \cos x dx$.**2**(d) Use partial fractions to find $\int \frac{3x + 1}{x^2 - x - 6} dx$.**3**

QUESTION THREE (13 marks) Use a separate writing booklet.

Marks

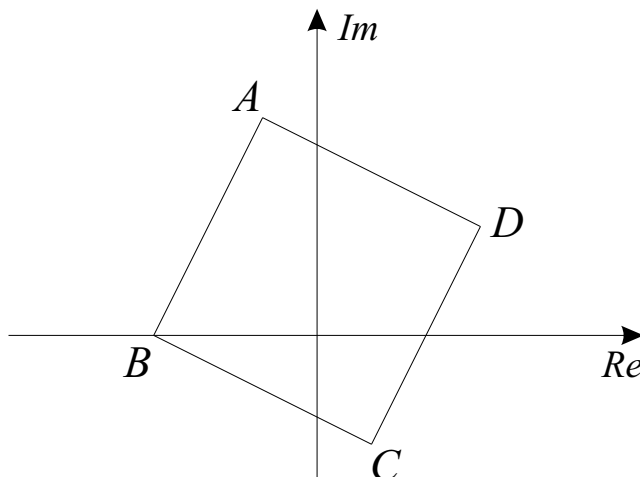
- (a) Use implicit differentiation to find the gradient of the tangent to the curve $x^2 - y^2 = 1$ at the point $(2, \sqrt{3})$. **2**
- (b) Let $z = p(\cos \alpha + i \sin \alpha)$ and $w = q(\cos \beta + i \sin \beta)$, where $p > 0$ and $q > 0$.
- (i) Find zw in the form $r(\cos \theta + i \sin \theta)$. **2**
- (ii) Hence prove that $|zw| = |z||w|$ and $\arg(zw) = \arg z + \arg w$. **1**
- (c) Let $z_1 = 1 - i$ and $z_2 = -\sqrt{6} + \sqrt{2}i$.
- (i) Express z_1 and z_2 in modulus–argument form. **4**
- (ii) Show that $z_1 z_2 = 4 \operatorname{cis} \frac{7\pi}{12}$. **1**
- (iii) Plot the point representing $z_1 z_2$ on the Argand diagram and hence find the exact value of $\tan \frac{7\pi}{12}$. **3**

QUESTION FOUR (13 marks) Use a separate writing booklet.**Marks**

- (a) Sketch the locus of
- z
- in the complex plane if
- $\arg(z - 1) = \frac{\pi}{4}$
- .

2

(b)



In the Argand diagram above, $ABCD$ is a square. A represents the number $z_1 = -1 + 4i$ and B represents the number $z_2 = -3 + 0i$.

- (i) Find the complex number represented by the vector $\overrightarrow{BA}$. **1**
- (ii) Hence find the complex number represented by the vector $\overrightarrow{BC}$. **1**
- (iii) Find the complex number represented by the point C . **1**
- (iv) Find the complex number represented by the point M , the intersection of the diagonals of $ABCD$. **1**
- (v) Let $w = \frac{3}{2}iz_1$ and let P represent the complex number w . Copy the diagram above and clearly show the point P . **1**

- (c) Evaluate
- $\int_0^{\frac{\pi}{2}} \sin^3 2x \, dx$
- .

3

- (d) Use the substitution
- $u = \sqrt{x}$
- to find
- $\int \frac{1}{(9+x)\sqrt{x}} \, dx$
- .

3

QUESTION FIVE (13 marks) Use a separate writing booklet.

Marks

- (a) Show that $\cos^{-1} \frac{3}{4} - \sin^{-1} \frac{9}{16} = \sin^{-1} \frac{1}{8}$. **3**
- (b) A complex number z satisfies $|z - 6i| = 3$. Use a diagram to find the maximum possible values of
- (i) $|z|$, **1**
- (ii) $\arg z$, where $-\pi < \arg z \leq \pi$. **2**
- (c) The complex numbers z_1 , z_2 , z_3 and z_4 correspond to the distinct points A , B , C and D respectively in the complex plane. If $z_1 - z_2 + z_3 - z_4 = 0$, what type of quadrilateral is $ABCD$? Justify your answer. **2**
- (d) Prove by mathematical induction that for all positive integers n , **5**
- $$\tan x + 2 \tan 2x + 4 \tan 4x + \cdots + 2^{n-1} \tan(2^{n-1}x) = \cot x - 2^n \cot(2^n x).$$

The Examination Continues over the Page

QUESTION SIX (13 marks) Use a separate writing booklet.

Marks

(a) Consider the function $f(x) = \tan^{-1}(\frac{1}{x})$.

(i) Describe the symmetry of the graph of $y = f(x)$.

1

(ii) Find $f'(x)$.

1

(iii) Describe the behaviour of $f(x)$ as $x \rightarrow 0$ from above and from below.

1

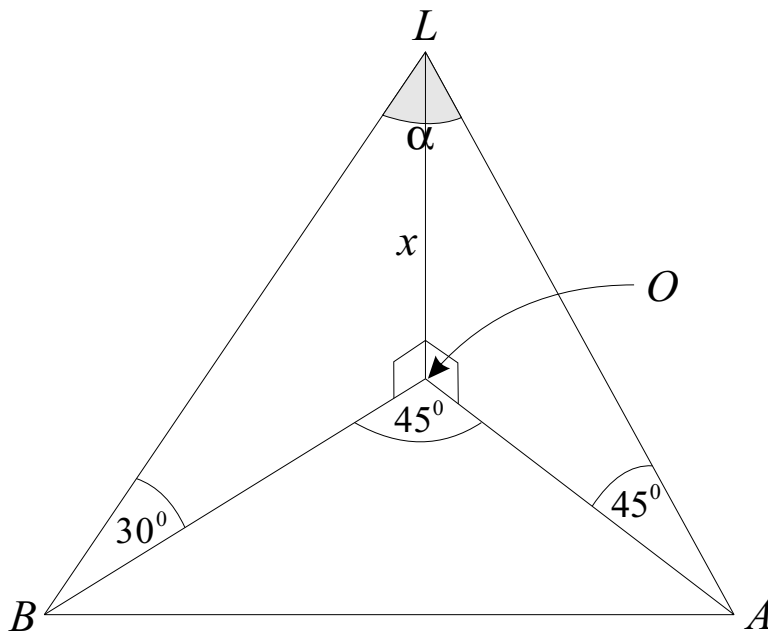
(iv) Find the equation of the horizontal asymptote.

1

(v) Sketch the graph of $y = f(x)$.

1

(b)



The diagram above shows a vertical light tower OL of height x metres. The angle of elevation of L from A is 45° , the angle of elevation of L from B is 30° and the angle AOB is 45° . Let $\angle ALB = \alpha$.

(i) Show that $AB^2 = (4 - \sqrt{6})x^2$.

2

(ii) Show that $\cos \alpha = \frac{2 + \sqrt{6}}{4\sqrt{2}}$.

1

(iii) Suppose that Pablo walks in a straight line from B to O . Let the variable point P represent his position and let the angle of elevation of L from P be θ .

2

Let $\angle ALP = \beta$ and show that $\cos \beta = \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta$.

(iv) If the light tower is 10 metres high, find the exact distance Pablo has to walk along BO so that his position P will minimise β . Justify your answer.

3

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2011 HALF-YEARLY EXAMINATION - EXTENSION 2

Question 1

(a) $z^2 + 4z + 5 = 0$

$$(z+2)^2 = -1$$

$$z = -2 + i \quad \text{or} \quad -2 - i$$

(b) $z = 2 - i, w = 3 - 4i$

(i) $iz = i(2 - i)$
 $= 1 + 2i$

(ii) $\overline{z+w} = \overline{5-5i}$
 $= 5+5i$

(iii) $\frac{1}{w} = \frac{1}{3-4i} \times \frac{3+4i}{3+4i}$
 $= \frac{3+4i}{9+16}$
 $= \frac{3}{25} + \frac{4}{25}i$

(iv) $(z-1)^2 = (1-i)^2$
 $= 1 - 2i - 1$
 $= -2i$

(c) substituting $z = 5 + 2i$, $25 + 20i - 4 - 50 - 20i + n = 0$
 $n = 29$

(d) (i) let $(x+iy)^2 = -24 + 10i$
 $x^2 - y^2 + 2ixy = -24 + 10i$

Equating real and imaginary parts:

$$\left. \begin{array}{l} x^2 - y^2 = -24 \\ xy = 5 \end{array} \right\} \begin{array}{l} x=1, y=5 \\ \text{or } x=-1, y=-5 \end{array}$$

The complex square roots are $1+5i$ and $-1-5i$.

(ii) $z^2 - (3+i)z + 8-i = 0$

$$\Delta = 9 + 6i - 1 - 32 + 4i$$

$$= -24 + 10i$$

$$z = \frac{3+i \pm (1+5i)}{2} \quad \text{from (i)}$$

$$z = 2+3i \quad \text{or} \quad 1-2i$$

Question 2

$$(a) \quad (i) \quad \int \frac{x^2}{(x^3+1)^2} dx = \frac{1}{3} \int 3x^2 (x^3+1)^{-2} dx$$
$$= -\frac{1}{3(x^3+1)} + C \quad \checkmark \checkmark$$

$$(ii) \quad \int \frac{1}{\sqrt{16-9x^2}} dx = \int \frac{1}{3\sqrt{\left(\frac{4}{3}\right)^2 - x^2}} dx$$
$$= \frac{1}{3} \sin^{-1} \frac{3x}{4} + C \quad \checkmark \checkmark$$

$$(b) \quad (i) \quad \int_0^{\frac{\pi}{6}} \sec 2x \tan 2x dx = \frac{1}{2} \left[\sec 2x \right]_0^{\frac{\pi}{6}}$$
$$= \frac{1}{2} (\sec \frac{\pi}{3} - \sec 0)$$
$$= \frac{1}{2} (2 - 1)$$
$$= \frac{1}{2} \quad \checkmark$$

$$(ii) \quad \int_{\frac{\pi}{3}}^{\pi} \frac{\sin x}{1 - \cos x} dx = \left[\ln |1 - \cos x| \right]_{\frac{\pi}{3}}^{\pi}$$
$$= \ln(1 - \cos \pi) - \ln(1 - \cos \frac{\pi}{3})$$
$$= \ln 2 - \ln \frac{1}{2}$$
$$= 2 \ln 2 \quad \checkmark$$

$$(c) \quad \int x \cos x dx = x \sin x - \int \sin x \times 1 dx \quad \checkmark$$
$$= x \sin x + \cos x + C \quad \checkmark$$

$$(d) \quad \text{let } \frac{3x+1}{(x-3)(x+2)} = \frac{A}{x+2} + \frac{B}{x-3}$$

$$3x+1 = A(x-3) + B(x+2) \quad \checkmark$$

$$\text{when } x = -2, \quad -5 = -5A$$

$$A = 1$$

$$\text{when } x = 3, \quad 10 = 5B$$

$$B = 2 \quad \checkmark$$

$$\int \frac{3x+1}{(x-3)(x+2)} dx = \int \frac{1}{x+2} + \frac{2}{x-3} dx$$
$$= \ln |x+2| + 2 \ln |x-3| + C \quad \checkmark$$
$$= \ln |x+2| (x-3)^2 + C$$

(a)

$$x^2 - y^2 = 1$$

$$2x - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{y}$$

At $(2, \sqrt{3})$

$$\text{gradient of tangent} = \frac{2}{\sqrt{3}}$$

(b)

$$z = p(\cos \alpha + i \sin \alpha), \quad w = q(\cos \beta + i \sin \beta)$$

$$\begin{aligned} \text{(i)} \quad zw &= pq(\cos \alpha \cos \beta + i \cos \alpha \sin \beta + i \sin \alpha \cos \beta - \sin \alpha \sin \beta) \\ &= pq((\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta)) \\ &= pq(\cos(\alpha + \beta) + i \sin(\alpha + \beta)) \end{aligned}$$

$$\text{(ii)} \quad \text{From (i), } |zw| = pq$$

$$= |z||w|$$

$$\text{and } \arg(zw) = \alpha + \beta$$

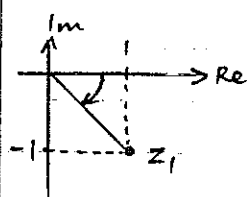
$$= \arg z + \arg w$$

(c)

$$\text{(i)} \quad z_1 = 1 - i$$

$$|z_1| = \sqrt{1+1}$$

$$= \sqrt{2}$$



$$\arg z_1 = -\frac{\pi}{4}$$

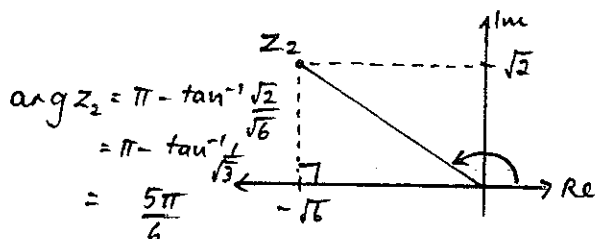
$$z_1 = \sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$$

(accept $\frac{7\pi}{4}$)

$$z_2 = -\sqrt{6} + \sqrt{2}i$$

$$|z_2| = \sqrt{6+2}$$

$$= 2\sqrt{2}$$



$$\arg z_2 = \pi - \tan^{-1} \frac{\sqrt{2}}{\sqrt{6}}$$

$$= \pi - \tan^{-1} \frac{1}{\sqrt{3}}$$

$$= \frac{5\pi}{6}$$

$$z_2 = 2\sqrt{2} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

(ii)

$$\begin{aligned} z_1 z_2 &= \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \times 2\sqrt{2} \operatorname{cis}\left(\frac{5\pi}{6}\right) \\ &= (\sqrt{2} \times 2\sqrt{2}) \operatorname{cis}\left(-\frac{\pi}{4} + \frac{5\pi}{6}\right) \\ &= 4 \operatorname{cis} \frac{7\pi}{12} \end{aligned}$$

$$\text{(iii)} \quad z_1 z_2 = (1 - i)(-\sqrt{6} + \sqrt{2}i)$$

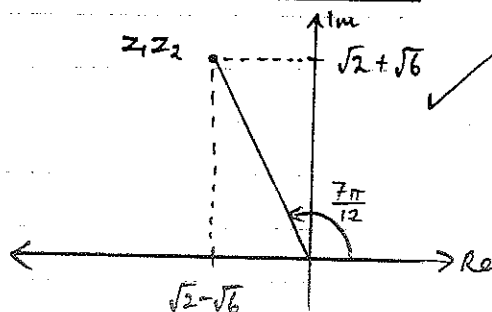
$$= -\sqrt{6} + \sqrt{2}i + \sqrt{6}i - \sqrt{2}i^2$$

$$= (\sqrt{2} - \sqrt{6}) + (\sqrt{2} + \sqrt{6})i$$

$$\tan \frac{7\pi}{12} = \frac{\sqrt{2} + \sqrt{6}}{\sqrt{2} - \sqrt{6}}$$

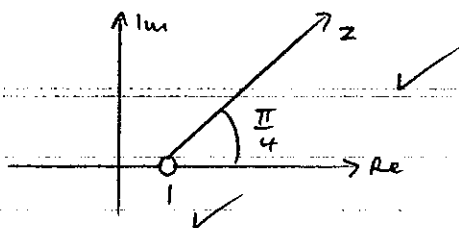
$$= -2 - \sqrt{3}$$

NOTE: $\sqrt{2} - \sqrt{6} < 0$



Question 4

(a)



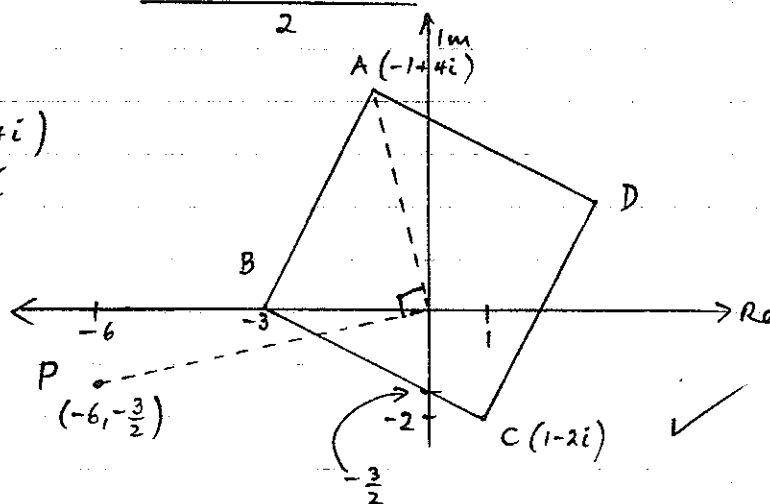
(b) (i) $\vec{BA}$ represents $z_1 - z_2 = -1 + 4i + 3$
 $= 2 + 4i$ ✓

(ii) $\vec{BC}$ represents $(2 + 4i)x - i = 4 - 2i$ ✓

(iii) $\vec{OC} = \vec{OB} + \vec{BC}$
 $= -3 + 4 - 2i$ So C represents $1 - 2i$.
 $= 1 - 2i$

(iv) M is the midpoint of AC:
M represents $\frac{-1 + 4i + 1 - 2i}{2} = i$ ✓

(v) $w = \frac{3}{2} i z_1$
 $= \frac{3}{2} i (-1 + 4i)$
 $= -6 - \frac{3}{2} i$



(c)

$$\begin{aligned} \int_0^{\pi/2} \sin^3 2x \, dx &= \int_0^{\pi/2} \sin 2x (1 - \cos^2 2x) \, dx \\ &= \int_0^{\pi/2} \sin 2x - \sin 2x \cos^2 2x \, dx \\ &= \left[-\frac{1}{2} \cos 2x + \frac{1}{6} \cos^3 2x \right]_0^{\pi/2} \\ &= \frac{1}{2} - \frac{1}{6} - \left(-\frac{1}{2} + \frac{1}{6} \right) \\ &= \frac{2}{3} \end{aligned}$$

(d)

Let $u = \sqrt{x}$
 $du = \frac{1}{2\sqrt{x}} \, dx$
 $2du = \frac{1}{\sqrt{x}} \, dx$ ✓

$$\begin{aligned} \int \frac{1}{(9+x)\sqrt{x}} \, dx &= 2 \int \frac{1}{3^2 + u^2} \, du \\ &= \frac{2}{3} \tan^{-1} \frac{u}{3} + C \\ &= \frac{2}{3} \tan^{-1} \frac{\sqrt{x}}{3} + C \end{aligned}$$

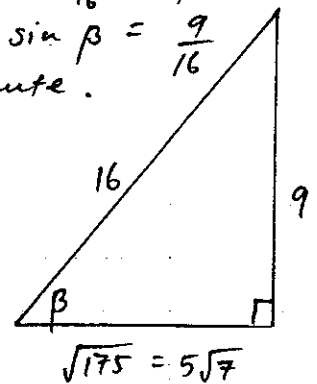
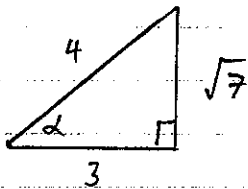
Question 5

(a)

$$\text{Let } \cos^{-1} \frac{3}{4} = \alpha \quad \text{and} \quad \sin^{-1} \frac{9}{16} = \beta$$

$$\cos \alpha = \frac{3}{4} \quad \sin \beta = \frac{9}{16}$$

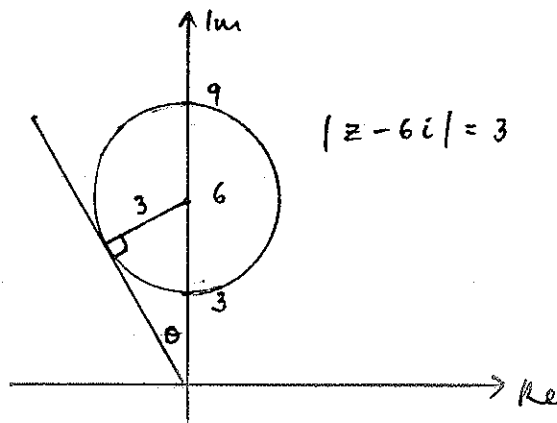
where α, β are acute.



$$\begin{aligned} \sin(\cos^{-1} \frac{3}{4} - \sin^{-1} \frac{9}{16}) &= \sin(\alpha - \beta) \\ &= \sin \alpha \cos \beta - \sin \beta \cos \alpha \\ &= \frac{\sqrt{7}}{4} \times \frac{5\sqrt{7}}{16} - \frac{9}{16} \times \frac{3}{4} \\ &= \frac{35 - 27}{64} \\ &= \frac{1}{8} \end{aligned}$$

$$\therefore \cos^{-1} \frac{3}{4} - \sin^{-1} \frac{9}{16} = \sin^{-1} \frac{1}{8}, \text{ since } (\alpha - \beta) \text{ is acute.}$$

(b)



(i) $\max |z| = 9$ ✓

(ii) $\sin \theta = \frac{3}{6}$
 $\theta = \frac{\pi}{6}$ ✓

$$\begin{aligned} \max \arg z &= \frac{\pi}{2} + \frac{\pi}{6} \\ &= \frac{2\pi}{3} \quad \checkmark \end{aligned}$$

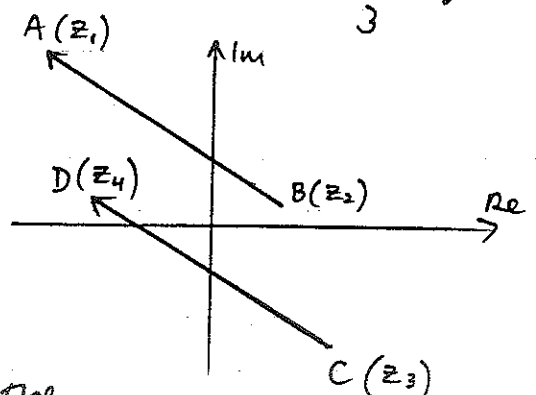
(c)

$$\begin{aligned} z_1 - z_2 + z_3 - z_4 &= 0 \\ z_1 - z_2 &= z_4 - z_3 \end{aligned}$$

So $z_1 - z_2$ represents $\vec{BA}$ ✓
 and $z_4 - z_3$ represents $\vec{CD}$.

Equal vectors ($\vec{BA} = \vec{CD}$) have the same magnitude and direction,

So ABCD is a parallelogram ✓
 (2 opposite sides equal and parallel).



Question 5 (cont.)

(d) $\tan x + 2 \tan 2x + 4 \tan 4x + \dots + 2^{n-1} \tan(2^{n-1}x) = \cot x - 2^n \cot(2^n x)$

A. when $n=1$, $LHS = \tan x$
 $RHS = \cot x - 2 \cot 2x$ ✓
 $= \frac{1}{\tan x} - 2 \times \frac{1 - \tan^2 x}{2 \tan x}$
 $= \frac{1 - 1 + \tan^2 x}{\tan x}$ ✓
 $= \tan x$
 $= LHS$

So the statement is true for $n=1$.

B. Assume the statement is true for some integer $n=k$.

ie Assume that

$$\tan x + 2 \tan 2x + 4 \tan 4x + \dots + 2^{k-1} \tan(2^{k-1}x) = \cot x - 2^k \cot(2^k x)$$

Let us prove the statement is true for $n=k+1$. ①

ie Prove

$$\tan x + 2 \tan 2x + 4 \tan 4x + \dots + 2^{k-1} \tan(2^{k-1}x) + 2^k \tan(2^k x) = \cot x - 2^{k+1} \cot(2^{k+1} x)$$
 ✓

$$\begin{aligned} LHS &= \cot x - 2^k \cot(2^k x) + 2^k \tan(2^k x) \text{ --- from ① } \checkmark \\ &= \cot x - 2^k (\cot 2^k x - \tan(2^k x)) \\ &= \cot x - 2^k \left(\frac{1}{\tan(2^k x)} - \tan(2^k x) \right) \\ &= \cot x - 2^k \left(\frac{1 - \tan^2(2^k x)}{\tan(2^k x)} \right) \\ &= \cot x - 2^k \times 2 \left(\frac{1 - \tan^2(2^k x)}{2 \tan(2^k x)} \right) \\ &= \cot x - 2^{k+1} \left(\frac{1}{\tan(2 \times 2^k x)} \right) \checkmark \\ &= \cot x - 2^{k+1} \cot(2^{k+1} x) \\ &= RHS \text{ as required} \end{aligned}$$

C. It follows, from parts A and B, by mathematical induction that the statement is true for all positive integers n .

Question 6

(a)

$$f(x) = \tan^{-1}\left(\frac{1}{x}\right)$$

$$\begin{aligned} (i) \quad f(-x) &= \tan^{-1}\left(-\frac{1}{x}\right) \\ &= -\tan^{-1}\left(\frac{1}{x}\right) \\ &= -f(x) \end{aligned}$$

i.e. $f(x)$ is an odd function.

So the graph of $y = f(x)$ has odd symmetry. ✓
i.e. Rotational symmetry of 180° about the origin.

$$\begin{aligned} (ii) \quad f'(x) &= \frac{-\frac{1}{x^2}}{1 + \frac{1}{x^2}} \\ &= -\frac{1}{1+x^2} \quad \checkmark \end{aligned}$$

(iii) Discontinuity at $x=0$.

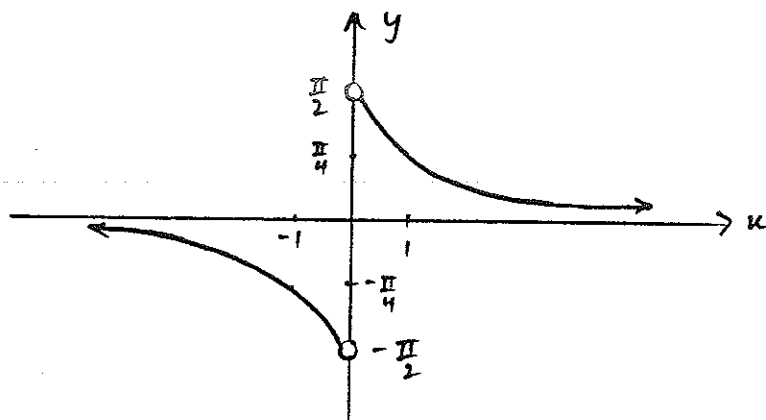
As $x \rightarrow 0^+$, $\frac{1}{x} \rightarrow \infty$, so $f(x) \rightarrow \frac{\pi}{2}$.

As $x \rightarrow 0^-$, $\frac{1}{x} \rightarrow -\infty$, so $f(x) \rightarrow -\frac{\pi}{2}$. ✓

(iv) as $x \rightarrow \infty$, $\frac{1}{x} \rightarrow 0$, so $f(x) \rightarrow 0^+$
as $x \rightarrow -\infty$, $\frac{1}{x} \rightarrow 0$, so $f(x) \rightarrow 0^-$

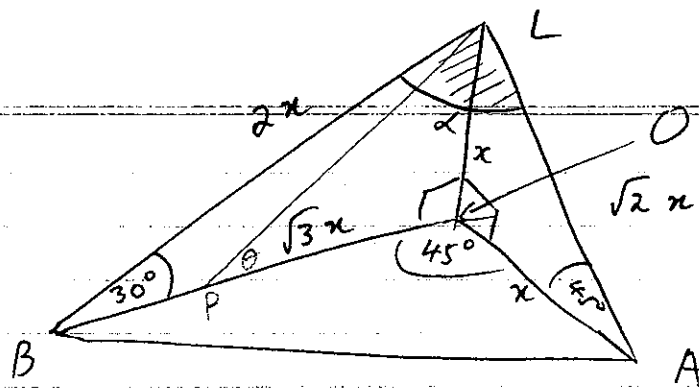
The horizontal asymptote has equation $y=0$. ✓

(v)



NOTE: $\frac{d}{dx}\left(\tan^{-1}\frac{1}{x}\right) = \frac{d}{dx}\left(-\tan^{-1}x\right)$

6 (b)



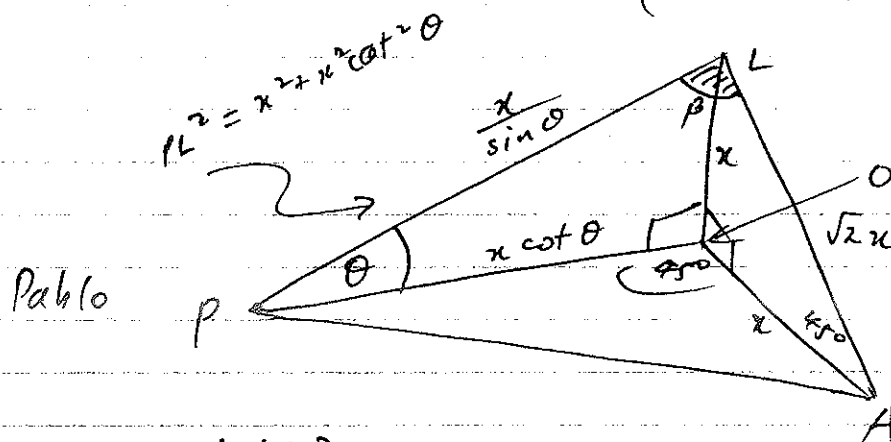
Right Triangle
Try to solve for
OA, OB, AL and BL.

$$\begin{aligned}
 (i) \text{ In } \triangle AOB, \quad AB^2 &= 3x^2 + x^2 - 2 \cdot \sqrt{3}x^2 \times \frac{1}{\sqrt{2}} \quad \checkmark \\
 &= 4x^2 - \sqrt{6}x^2 \quad \checkmark \\
 &= (4 - \sqrt{6})x^2
 \end{aligned}$$

(ii) From $\triangle ALB$,

$$\begin{aligned}
 AB^2: \quad 4x^2 + 2x^2 - 2 \cdot 2\sqrt{2}x^2 \cos \alpha &= (4 - \sqrt{6})x^2 \\
 6 - 4\sqrt{2} \cos \alpha &= 4 - \sqrt{6} \quad \checkmark \\
 \cos \alpha &= \frac{2 + \sqrt{6}}{4\sqrt{2}} \\
 (\alpha &\doteq 38^\circ)
 \end{aligned}$$

(iii)



$$AL^2 = x^2 + x^2 \cot^2 \theta$$

Pablo

as in (ii)

cos rule in $\triangle AOP$

cos rule in $\triangle ALP$

$$\begin{aligned}
 AP^2: \quad x^2 \cot^2 \theta + x^2 - 2x^2 \cot \theta \left(\frac{1}{\sqrt{2}} \right) &= 2x^2 + x^2 + x^2 \cot^2 \theta \quad \checkmark \\
 1 - \sqrt{2} \cot \theta &= 3 - 2\sqrt{2} \frac{\cos \beta}{\sin \theta} \\
 \boxed{x \sin \theta} \quad -\sqrt{2} \cos \theta &= 2 \sin \theta - 2\sqrt{2} \cos \beta \quad \checkmark \\
 \cos \beta &= \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta
 \end{aligned}$$

as required

(iv)

$$\cos \beta = \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta$$

$$\text{let } a \cos(\theta - u) = \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta$$

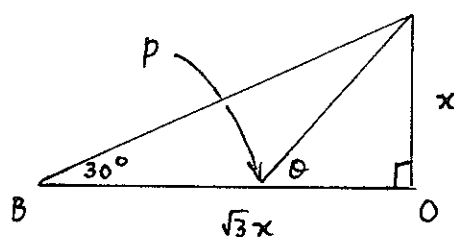
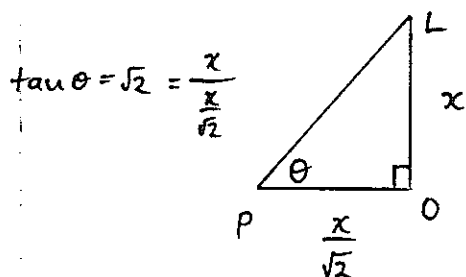
$$a \cos \theta \cos u + a \sin \theta \sin u = \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta$$

$$\therefore \left. \begin{array}{l} a \cos u = \frac{1}{2} \\ a \sin u = \frac{1}{\sqrt{2}} \end{array} \right\} \begin{array}{l} a^2 = \frac{1}{4} + \frac{1}{2} \\ a = \frac{\sqrt{3}}{2} \end{array} \quad \begin{array}{l} \tan u = \frac{1}{\sqrt{2}} \times \frac{2}{1} \\ u = \tan^{-1} \sqrt{2} \end{array}$$

$$\cos \beta = \frac{\sqrt{3}}{2} \cos(\theta - \tan^{-1} \sqrt{2})$$

$$\beta = \cos^{-1} \left(\frac{\sqrt{3}}{2} \cos(\theta - \tan^{-1} \sqrt{2}) \right)$$

β is minimised when $\cos(\theta - \tan^{-1} \sqrt{2})$ is maximised.
i.e. when $\theta = \tan^{-1} \sqrt{2}$



When $x=10$, $OP = \frac{10}{\sqrt{2}} = 5\sqrt{2}$

So when Pablo has walked $(10\sqrt{3} - 5\sqrt{2})$ m, β is at a minimum.

NOTE: Minimum β is 30° when $\theta = \tan^{-1} \sqrt{2} \doteq 55^\circ$.
When $\theta = 30^\circ$, $\beta \doteq 38^\circ$
When $\theta \rightarrow 90^\circ$, $\beta \rightarrow 45^\circ$.

OR Using Calculus :

$$(iv) \quad \beta = \cos^{-1} \left(\frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta \right)$$

$$\frac{d\beta}{d\theta} = - \frac{\frac{1}{\sqrt{2}} \cos \theta - \frac{1}{2} \sin \theta}{\sqrt{1 - \left(\frac{1}{2} \sin \theta + \frac{1}{\sqrt{2}} \cos \theta \right)^2}}$$

$$\text{when } \frac{d\beta}{d\theta} = 0, \quad \frac{1}{2} \sin \theta = \frac{1}{\sqrt{2}} \cos \theta$$
$$\tan \theta = \sqrt{2}$$
$$\theta = \tan^{-1} \sqrt{2} \approx 55^\circ$$

Implicit Differentiation

$$\cos \beta = \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta$$

$$-\sin \beta \frac{d\beta}{d\theta} = \frac{1}{\sqrt{2}} \cos \theta - \frac{1}{2} \sin \theta$$

$$\text{So } \frac{d\beta}{d\theta} = 0 \text{ when}$$

$$\frac{1}{2} \sin \theta = \frac{1}{\sqrt{2}} \cos \theta$$

since $\sin \beta > 0$

$$\text{ie } \tan \theta = \sqrt{2}$$

θ	50°	$\tan^{-1} \sqrt{2}$	60°
$\frac{d\beta}{d\theta}$	-0.101	0	0.109
	(3dp)		(3dp)

So we have minimum β when $\theta = \tan^{-1} \sqrt{2}$.

The result follows...

OR To minimise β we want to maximise $\cos \beta$.

$$\text{Let } \psi = \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{2} \cos \theta$$

$$\frac{d\psi}{d\theta} = \frac{1}{\sqrt{2}} \cos \theta - \frac{1}{2} \sin \theta$$

$$\text{when } \frac{d\psi}{d\theta} = 0, \quad \frac{1}{\sqrt{2}} \cos \theta = \frac{1}{2} \sin \theta$$
$$\tan \theta = \sqrt{2}$$

$$\frac{d^2\psi}{d\theta^2} = -\frac{1}{\sqrt{2}} \sin \theta - \frac{1}{2} \cos \theta$$

$$\frac{d^2\psi}{d\theta^2} < 0 \text{ for } \theta = \tan^{-1} \sqrt{2}$$

So $\cos \beta$ is maximised for $\theta = \tan^{-1} \sqrt{2}$.

The result follows...